# Evidence-Bound Harness Evolution: Search, Failure Memory, and Held-Out Transfer

Jeremy H. Carroll

2026-10-23

**Code baseline:** 59dde95a7860063c25d75db37f40af556e9f28ee

## Abstract

An agent can propose changes to the software that organizes its own attempts. Whether the changes improve research performance is an experimental question; whether a particular submitted result passes a declared checker is a separate verification question. We implement an explicit experimental re-entry path over existing Omega Engine subsystems, inspired by mechanisms reported for AIDE2. The path retains candidate ancestry, isolates candidate execution, reserves provider liability against one durable budget, separates public feedback from private grading, and requires fresh confirmation before replacing an experimental incumbent. A frozen protocol allocates a maximum of $40 to three evolution trajectories, untouched transfer, ablations and an exploratory ignition comparison. This revision contains implementation and offline checks. Linux testing and the paid campaign were deferred by the operator on 23 October 2026, so there are no new model trials, measured improvements or transfer results. Missing experiments remain explicitly unmeasured. Neither the implementation nor an eventual favorable trajectory would establish accelerating self-improvement.

## 1. Question and source boundary

The question is whether changes to a bounded research harness improve exact task completion, cost, or both under a fixed provider and evaluator configuration. The experimental object is the harness: its prompts, workflow options, search policy and context policy. The task checker and the budget controller are parts of the measuring apparatus. Giving the experimental object write access to that apparatus would change the meaning of an observed improvement.

Srikanth and colleagues describe a nested optimization in which an outer agent rewrites an inner research agent and evaluates each rewrite on hidden grades [1, pp. 3–4]. They report seven accepted improvements during an eight-day run and evaluate transfer on additional benchmarks [1, pp. 5–7]. These are the authors' reported findings. They are not observations from this repository, independent replications, or premises of the finite PFP theorems.

The supplied source is arXiv:2609.26457v1, 28 pages, with SHA-256 8ba79e0ad1028e43b0a47e e06162c9b3996ec90f7990b7ec35993444e505409b. The source-to-claim map identifies page-level support and separates reported findings, implemented mechanisms, measurements and hypotheses. The version and digest are fixed even if a later revision of the external paper appears.

## 2. Mechanism inventory and ownership

| Owner | Implemented responsibility | Evidence ceiling |
|---|---|---|
| OEOP | Versioned experiment, candidate, evaluation and decision contracts; existing typed workflow compilation | A typed connection does not prove callback correctness |
| Woland | Explicit research orchestration, public search, private evaluation, confirmation and retained evidence | Experimental selection does not deploy a capability |
| PFP | Exclusive provider admission, transport, reservations and token-usage settlement | List-price accounting is not an invoice audit |
| OEEIS | Scoped failure guidance with reproducers and benign controls | Diagnostic candidates are not admitted or activated antibodies |
| OEPCIV | Protected host context, candidate genealogy, rejected branches and exact Pareto comparison | Ancestry does not authenticate an external author |
| OEOS | Rootless container process and resource control | Isolation must be observed on the actual Linux host |

The command exposes `preflight`, `run`, `resume`, `replay` and `report`. Ordinary learning retains its recursion suppression and publication rules. The research use case can continue from confirmed candidate bytes without opening a pull request or granting production admission. This is an explicit experimental use case, not a new general agent framework.

Typed workflow tasks reuse the C/R/E/I/M/T repair sequence described in Paper 028. Its host-selected check callback invokes the container adapter and checks the downstream output. The earlier supplied-patch example remains historical evidence of a narrower path. Working-copy separation in that example is not retroactively described as an operating-system sandbox.

## 3. Candidate and evidence identities

An experiment fixes source and checker manifests, all task partitions, model IDs and configuration, the original budget identity, and analysis choices. A candidate retains its parent, source identity, strategy label, mutation description and complete allowed file bytes. The only editable files are `prompt.txt`, workflow.json, search_policy.py and context_policy.py. The serialized editable file set is capped at 3,600 bytes to reserve room for outer re-entry context. The outer role receives the root, current champion, six task aggregates and two recent decision references from its own trajectory. Private fixtures, checkers, provider credentials, admission and accounting code are outside that surface.

An evaluation occurrence identifies the experiment, candidate, parent, source, checker, split, task, repetition, model and configuration. It retains the submitted program, raw model response references, actual execution outputs, public and private verdicts, and cost. An incumbent decision cites the selection and confirmation receipts on which it depends. Changed candidate bytes, a

stale parent, a substituted checker or a foreign task cannot inherit a prior acceptance merely by preserving a favorable label.

Content-addressed objects and a chained journal detect inconsistent local alterations. These checks do not authenticate a provider independently of the trusted transport, protect against a trusted host rewriting all records and anchors, or establish a reproducible compiler build. Provider repetitions are fresh calls; replaying one response against a checker is not another trial.

## 4. Reference search and bounded context

The reference has five strategy labels: conservative, aggressive rewrite, ensemble, tuned specialist and robust simple. After untried arms are sampled, UCB1 uses empirical mean public score and the exploration term $\sqrt{2 \log N / n_a}$. With probability 0.30 it instead samples a softmax over the best score seen for each arm, at temperature one. Every fifth step forks the champion under a different strategy. The local seed, random draw, selected arm, parent and scheduling constants are retained. These choices implement a specific reference policy inspired by [1, p. 9]; they are not identified as the authors' exact source code or a reproduction of their benchmark results.

The drafting and repair contexts include the root, champion, two recent attempts and retained evidence references. Host checks refuse a projection that drops required references. The reference compresses attempt detail into summaries; the full-history comparator retains prior code. Both remain subject to the same provider input bound. The host's conservative text-byte ceiling is stricter than the nominal 8,000-input-token allowance.

Failure memory is considered only when candidate execution errors reach 15% of relevant observat ions. At most three deduplicated signatures are included. Provider errors are a separate category. Each eligible signature retains a reproducer and benign control and remains scoped to the task kind. These thresholds are design choices from [1, p. 9], not calibrated constants of a general reliability theory.

A repair must include a diagnosis when the diagnosis mechanism is enabled. The host verifies its presence and retains it; it does not certify the truth of a natural-language root-cause explanation. A substantive-output gate rejects empty or nearly empty programs before execution. Exact downstream checks still determine task success. The ineffective median-score adjustment discussed in [1, p. 10] is deliberately omitted.

## 5. Frozen tasks and comparators

Selection uses two instances from each of three families: exact-type and stale input binding repairs, C052 `max-flow/min-cut` and C054 linear-system certificates, and typed workflows with checked downstream results. Fresh confirmation cases are frozen separately; both selection and confirmation remain optimization data. Public input projection omits expected outputs, reference implementations and private fixture databases. Identifiers do not encode the desired verdict.

Untouched transfer contains four new tasks from each selection family and eight C059 SQL/NULL-`semantics` tasks. The SQL family is excluded from discovery. Cases are generated afresh from the fixed local generation seed. Private inputs and grading results are never injected into the inner search context. Private selection feedback may inform later outer rewrites, so it is not described as held-out evidence after optimization.

The current-policy comparator projects the existing sequential `propose/check` behavior into the

common task adapter. It has no strategy diversification, compression, failure memory or diagnosis requirement. It is not a claim that the unchanged production learning service has run this campaign. The second baseline is the hand-engineered reference. Three independent trajectories start from the same reference and each allow six outer rewrite proposals. The final incumbent of every trajectory is retained, including a reference that never improves.

Each discovery comparison uses paired candidate/incumbent evaluations, three fresh repetitions per task and at most six inner turns. Exact completion fractions and cost remain separate objectives. The ordering is correctness first, cost second, with the incumbent retained on unresolved ties. Fresh confirmation, no task-family regression, and invariant controls are necessary for replacement. Failed, refused, timed-out and incomplete assignments remain in the denominator.

## 6. Budget, restart and execution isolation

| Phase | Maximum USD |
|---|---|
| Preflight | 2 |
| Discovery and confirmation | 16 |
| Untouched transfer | 10 |
| Ablations | 6 |
| Exploratory ignition | 4 |
| Infrastructure retry reserve | 2 |
| Total | 40 |

The outer model is claude-sonnet-5-5 and the inner model is claude-haiku-5-5. Requests use explicit adaptive-thinking configurations, no sampling parameters unsupported by those models, no tools and no prompt caching. Inner output is capped at 2,000 tokens and outer output at 3,000. The same SQLite budget persists across phases, descendants and restarts. Atomic reservations enforce phase limits and at most two in-flight requests; the initial dispatcher is sequential. An interrupted call or missing usage retains its maximum liability and prevents further dispatch pending reconciliation. Rejected response content can still carry billable usage.

At the retained 2026-10-23 uncached observations, a maximally sized inner call reserves $0.0018 and an outer call $0.046. The planned discovery bound of 8,100 inner calls and 18 rewrites is $15.408 before retries. This is a planning calculation, not measured expenditure. Pricing observations expire and must be refreshed before dispatch. Infrastructure hosting is outside the API ceiling.

The prepared deployment targets a user-provisioned Ubuntu 24.04 x86-64 server with rootless Podman and cgroup v2. Candidate containers have no network, credentials, private evaluator mounts or repository write access. They have read-only source mounts, bounded memory, CPU, process count, file size, output and elapsed execution time. A real isolation probe is a prerequisite for paid discovery. This revision has not executed that probe on the requested server.

## 7. Analysis, ablations and ignition

Transfer analysis pairs harnesses by task and averages the three fresh model trials within each task. It reports exact completion fractions, token usage, list-price cost, `public/private` disagreement, execution errors, context size and retained strategy diversity. Confidence intervals use 2,000 paired task-level bootstrap draws with a declared local analysis seed. Comparative tests use a two-sided

exact paired sign test. Holm correction covers seven predeclared transfer contrasts: reference versus current policy and each final incumbent versus both baselines. Incomplete panels do not earn comparative claims.

A separate 12-task panel compares trajectory zero's final incumbent with four variants removing diversification, compression, failure memory and diagnosis separately. Choosing trajectory zero before transfer prevents selection on the transfer outcomes. The four ablation contrasts form a separate Holm family. These are tests of the implemented mechanisms in this bounded setting, not universal causal explanations for the external paper's gains.

The ignition experiment compares the original outer policy with the evolved outer harness in three paired repetitions of three rewrite steps. Both outer arms use Sonnet and the same initial inner reference. To remain within $4, ignition inner evaluations use one repetition and one inner turn per task, with separately assigned confirmation calls. The protocol records this lower resolution. Results are exploratory. The external paper likewise reports its own ignition comparison as inconclusive [1, pp. 10–11]. Neither experiment establishes accelerating returns or an indefinitely improving process.

## 8. Current results and remaining execution

The machine-readable campaign report is produced by offline replay. At this revision it records **NOT_RUN**, zero independent provider calls, zero assigned model trials, zero settled API spend and zero held API liability. Consequently, completion rates, transfer effects, ablation effects and ignition effects are **NOT_MEASURED**. No synthetic trace is presented as a live experiment.

Local verification exercises exact-type contracts, protected edit surfaces, confirmation and family-regression gates, memory scope, raw-byte tampering, concurrent reservations, restart holds and malformed-answer settlement. Existing repair and ordinary-learning tests are run separately. The reproduction guide identifies commands and the retained verification receipt. Windows unit checks do not certify Linux containment.

The frozen campaign must be replayed against its retained source bytes. Subsequent main changes are not silently substituted into that experiment. The reproduction guide supplies an archive-bound offline command that checks package custody both before and after replay. Its isolated Python process uses existing dependencies and refuses networking and child processes. This follow-up improves reproduction of the zero-call report; it supplies no additional model trial, Linux isolation observation or fresh dependency-installation result. The original review archive retains the earlier manuscript snapshot.

Linux testing and paid execution were initially deferred for this delivery. The operator subsequently selected Windows verification and removed Linux testing from scope. The Windows checks exercise offline behavior; the frozen rootless runner remains Linux-specific. They do not establish Windows containment for generated programs. A Windows live campaign would require a separately reviewed containment adapter and protocol revision. Provider compatibility, containment, live task calibration and the entire paid campaign remain NOT_RUN. An experiment curtailed by budget or infrastructure must retain its assigned trials and refusal records; it must not be rewritten as a successful trajectory.

## 9. Relation to finite PFP results

Paper 024's solution-map composition theorem requires explicit soundness and coverage hypotheses. Experimental incumbent selection supplies neither a new solution map nor proof of those hypotheses.

Shared fuel limits expenditure; it does not imply useful progress.

Paper 026's future-preservation result concerns a finite deterministic quotient preserving action availability, exact cost, successor classes and terminal labels. The heuristic context summaries here do not meet that specification merely by being shorter. Dropping a reference can disable a later operation; retaining a reference can still lose the content needed to use it correctly. Measured context size and task completion therefore remain distinct from operation-preservation claims.

The claim-scoping rule in Paper 023 applies separately to each checked output. The compatibility theorem in Paper 025 does not turn independently favorable task scores into a coherent global guarantee. Paper 027 permits changing the proposal channel while holding host obligations fixed. This study adds an implemented experimental procedure, not a new formal theorem or Lean result.

## References


1. Dhruv Srikanth, Bingchen Zhao, Dixing Xu, Yuxiang Wu and Zhengyao Jiang. *Recursive self-improvement of AI research agents.* arXiv:2609.26457v1, submitted 22 September 2026. Versioned source.
2. Paper 024: recursive re-entry and shared fuel.
3. Paper 026: future-faithful compression and cost.
4. Paper 028: typed composition and bounded DDC repair.
5. Anthropic, Haiku 5.5 model overview and Sonnet 5.5 model overview, observed 23 October 2026.